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Title

Design Aspects and Performance Analysis of Multicarrier Waveform contenders of 5G and beyond Wireless Networks

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Abstract

- Next generation wireless network are intended to support heterogeneous networks and to meet the major challenge of coping with spectral efficiency (SE) to accommodate larger number of users with reduced complexity and latency.
- The key challenge in selecting the modulation waveform is that it must be flexible, spectral efficient and compatible with the service required.
- In this paper various design parameters of multi carrier waveform: Filtered Orthogonal Frequency Division Multiplexing (F-OFDM), Universal Frequency Multi Carrier (UFMC), Filter Bank Multi Carrier (FBMC), and Generalized Frequency Division Multiplexing (GFDM) have been investigated and analyzed.

Introduction

5G has mainly three service scenarios [1]

Use Cases	Requirements
Ultra-Reliable Low-Latency Communication (uRLLC)	To support low-latency transmissions of small blocks with very high reliability from a limited set of terminals
Enhanced Mobile Broadband (eMBB)	To support stable connections with very high peak data rates, as well as moderate rates for cell-edge users
Massive Machine Type Communication (mMTC)	To support massive number of Internet of Things (IoT) device

Problem & Theories

- Problem statement: 5G cope with challenges like increase in user density, seamless connectivity, traffic density, data rate, and extensive applications.

$$\text{Area throughput [bit/s/km}^2 \text{]} = B \text{ [Hz]} \cdot D \text{ [cells/km}^2 \text{]} \cdot SE \text{ [bit/s/Hz/cell]}$$

■ Possible Solutions:

1. One potential solution would be to increase the bandwidth B
2. Another potential solution would be to densify the cellular network by deploying $1000\times$ more BSs per km^2

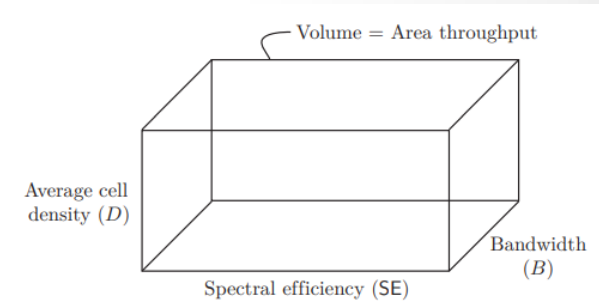


Fig.1. Three ways to improve the area throughput of cellular networks [2]

Problems in Above Solutions: Higher cell density and larger bandwidth are approaching at saturation point where further improvements are increasingly complicated and expensive.

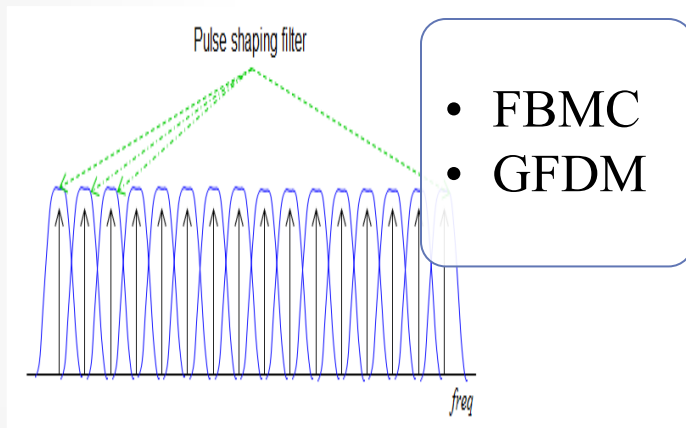
Ultimate Solution: Increasing the SE by virtue of new modulation and multiplexing techniques.

Focus on New Techniques

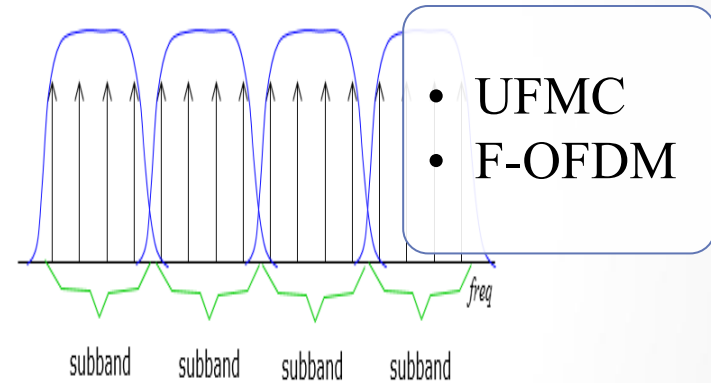
- With the objective of enhancing the spectrum efficiency and providing the diverse services, the design of air interface needs to be considered.
- Orthogonal Frequency Division Multiplexing (OFDM) is widely used air interface in fourth generation long term evolution (4G LTE) system.
- OFDM suffers from various shortcomings like high PAPR, carrier frequency offset (CFO), and loss of spectral efficiency due to insertion of cyclic prefix.
- The waveform to be implemented at the physical layer in 5G must be able to perform in high dense networks with heterogeneous traffic and high throughput.

Air Interface Methods [3]

Subcarrier wise Filtering



Sub-band wise Filtering



Filter Bank Multi Carrier (FBMC)

- In FBMC the data stream is converted in a set of parallel data and then transmitted through a bank of modulated filters.
- Filtering of the signal is done on a sub-carrier basis using a long prototype filter parameterized by the overlapping factor K .
- In FBMC each symbol overlaps with K neighboring symbols in the time domain.
- There is no need to insert the cyclic prefix (CP) which enhances spectral efficiency [4].

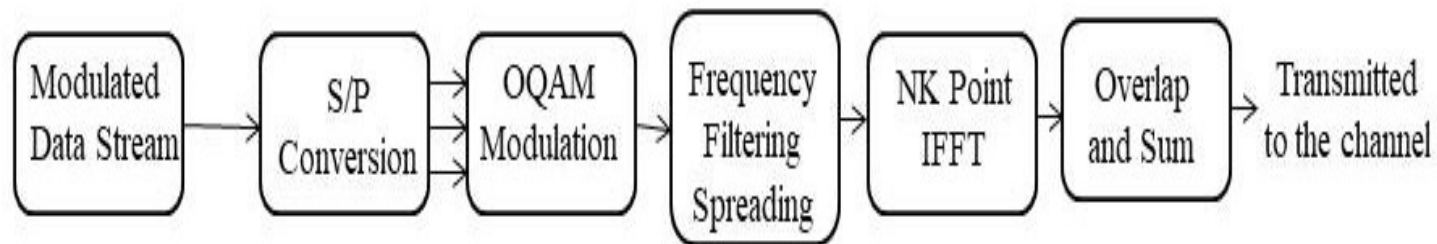


Figure 2 FBMC Transmitter

Generalized Frequency Division Multiplexing (GFDM)

- GFDM waveform is based on non-orthogonal sub carriers spread the data block in time-frequency domain.
- GFDM symbol is composed of M QAM symbols for each of the N sub-carriers resulting in transmission of $K = NM$ complex modulated data.
- The shaping filter used here is Root Raised Cosine filter (RRC) filter. To avoid ISI, a CP is added at the end of each block of symbols with constraint that CP length must be greater than delay spread[5].

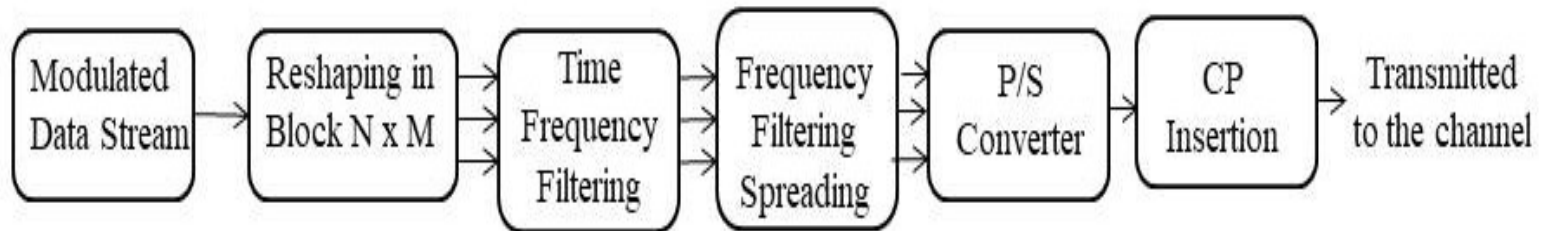


Figure 3 GFDM Transmitter [6]

Universal Filtered Multicarrier (UFMC)

- In this technique, the overall bandwidth is divided into sub-bands. UFMC filters the signal on a sub-band basis
- Filter at each sub-band is applied and here Dolph-Chebyshev prototype filter is used.

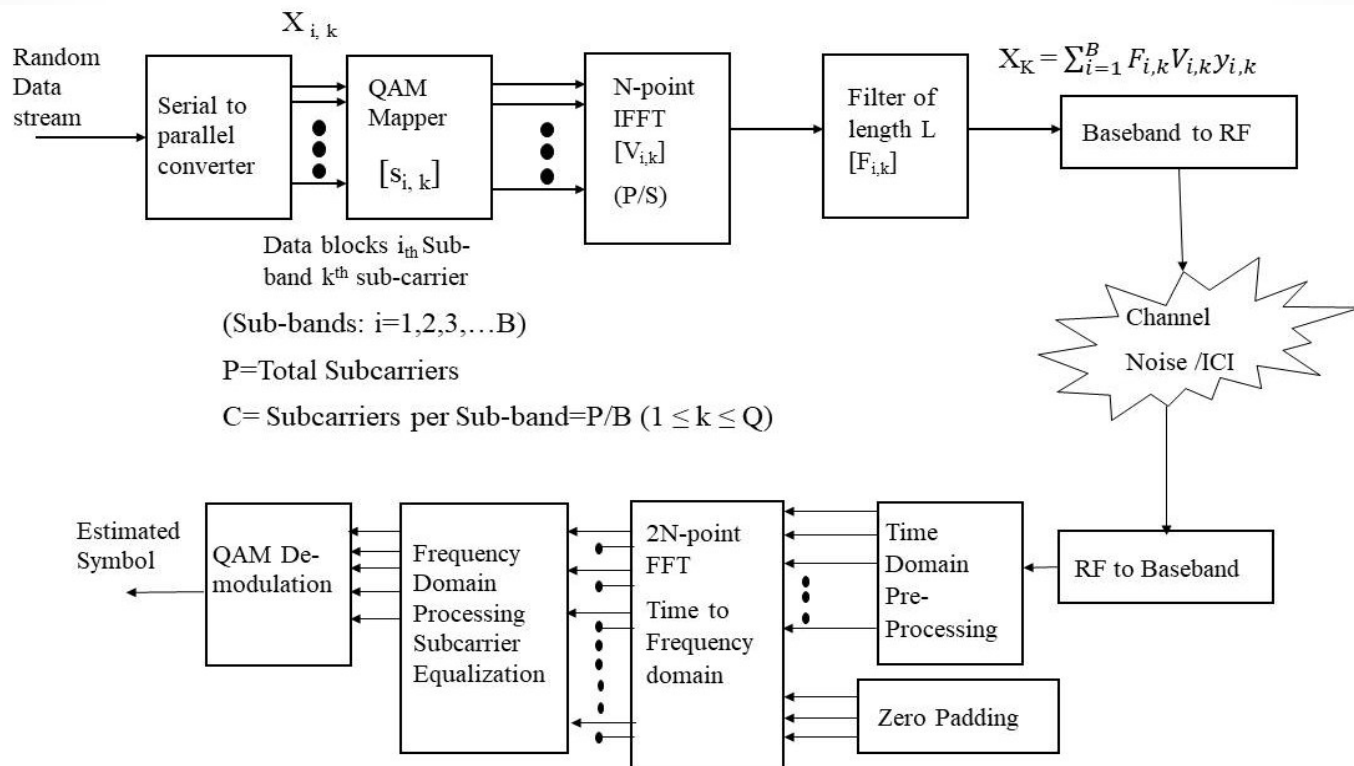
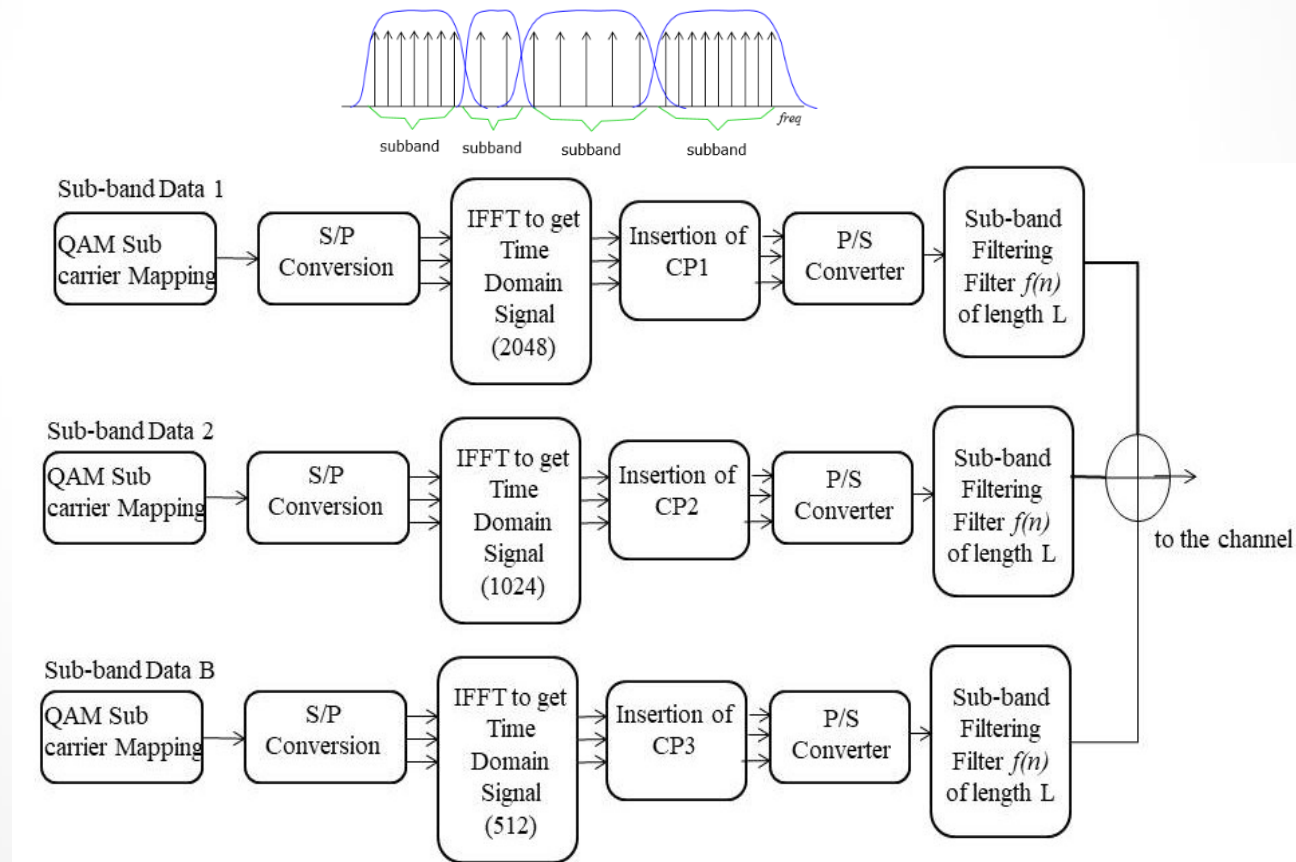


Figure 4 UFMC Transceiver [7]

Filtered OFDM (F-OFDM)

- In Filtered OFDM, the bandwidth is divided into sub-bands but each sub-band has variable length. The sub-band size is adjusted according to the different channel conditions in the time–frequency domain.
- This method provides flexibility to choose sub-carrier spacing and filter design but have very high implementation complexity



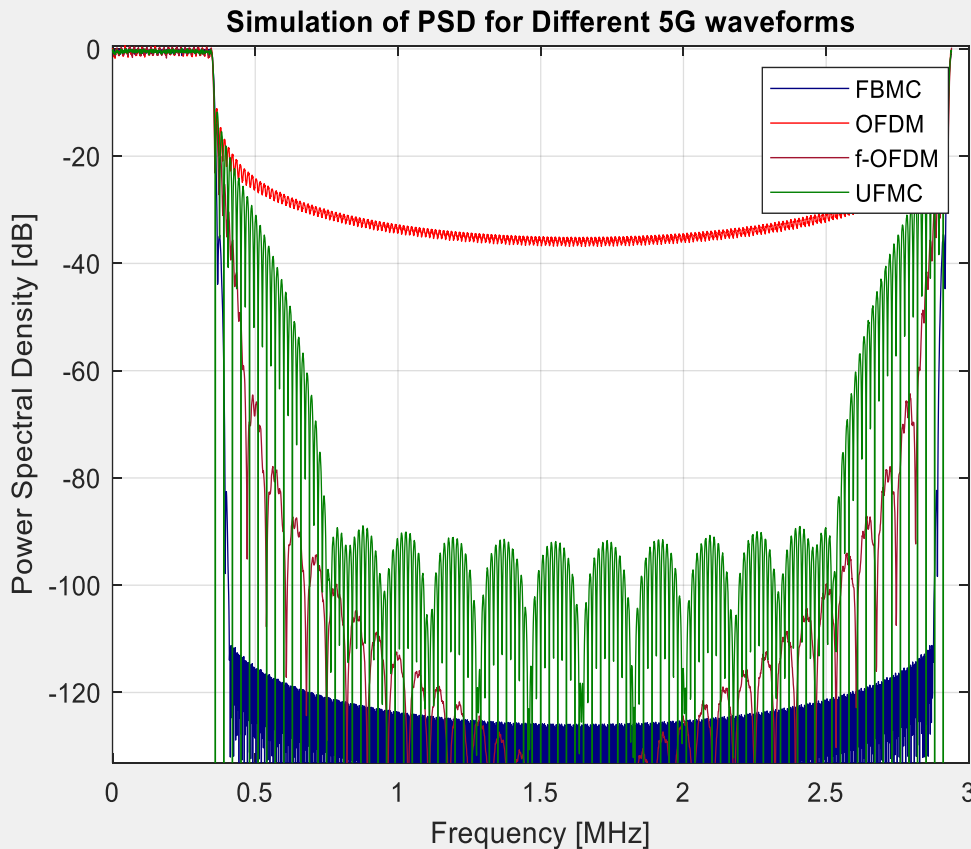
Simulation Parameters

To analyze the performance of waveforms and to investigate its influence, the system model is simulated in MATLAB 2022b

Waveform Technique	Simulation Parameter	Value
FBMC	Number of Sub carriers	12
	Overlapping factor K	4
	Modulation type	64 QAM
	Filter	PHYDYAS [9], Hermite
	Length of the weighting filter window	43
	Sub carrier spacing	15KHz
f-OFDM	Number of Sub-bands	10
	Modulation type	64 QAM
	Filter	Hanning window
GFDM	Block length	32
	Number of Sub carriers	128
	RRC filter Roll off	0.2
	Modulation type	16QAM
UFMC	Number of Sub-bands	10
	Sub band Size	20
	Sub-band Offset	156
	Modulation type	4, 16, 64, 256 QAM
	Size of IFFT	512
	Filter	Dolph-Chebyshev
	Length of the weighting filter window	43
	Sideband filter attenuation	40 dB

System Simulation and Results

1. PSD for the 5G candidate waveforms: FBMC, OFDM, f-OFDM, UFMC



Remarks:

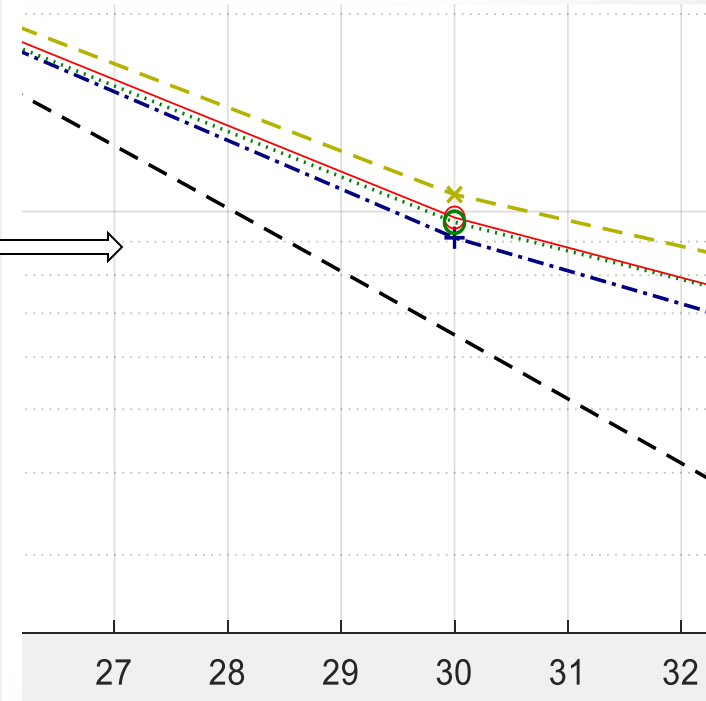
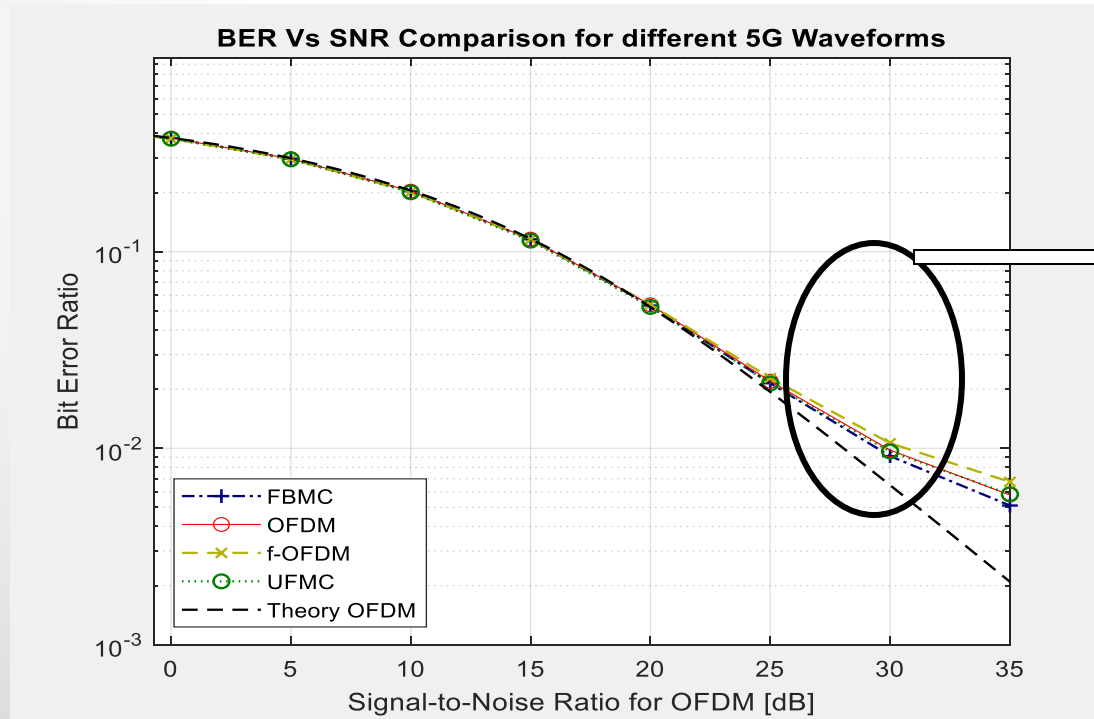
- ✓ FBMC is the most spectrally efficient waveform with minimum OOB.
- ✓ In FBMC sub-carrier in the system which weakens the inter-carrier-interference

System Simulation and Results

2. The comparison of a BER performance of all the 5G waveforms with respect to Signal to Noise Ratio (SNR) [10]

Remarks:

- ✓ f-OFDM is having higher BER as compared to other multi-carrier modulation techniques.
- ✓ In contrast to this UFMC achieves optimal BER performance.

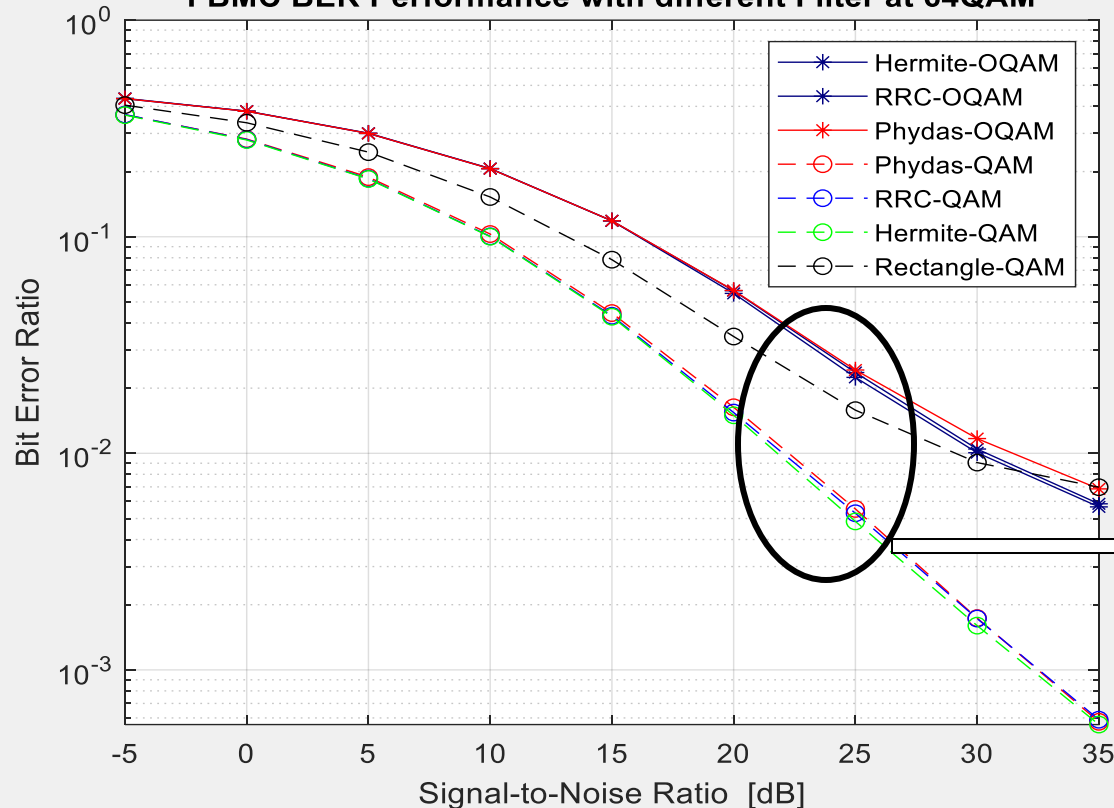


System Simulation and Results

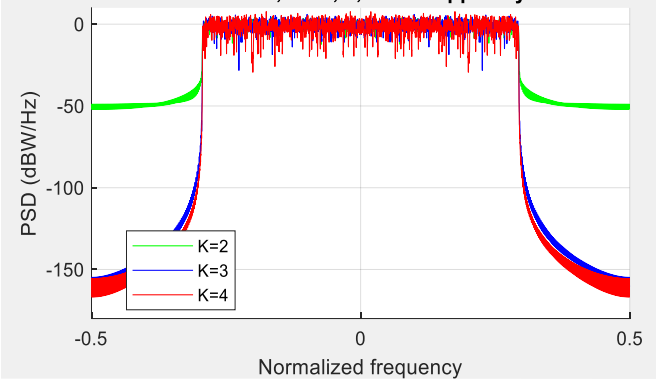
FBMC simulation

- ✓ PSD for FBMC with varying the value of K gives different filter windows [11].
- ✓ The analysis of FBMC with the metric of BER with different filters at 64QAM

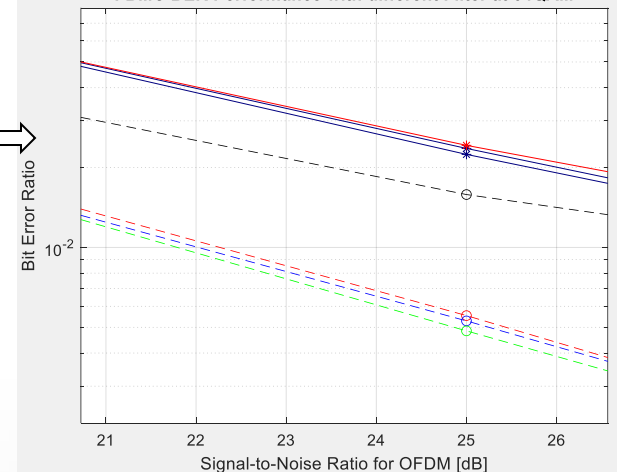
FBMC BER Performance with different Filter at 64QAM



PSD for FBMC, K =2, 3, 4 overlapped symbols



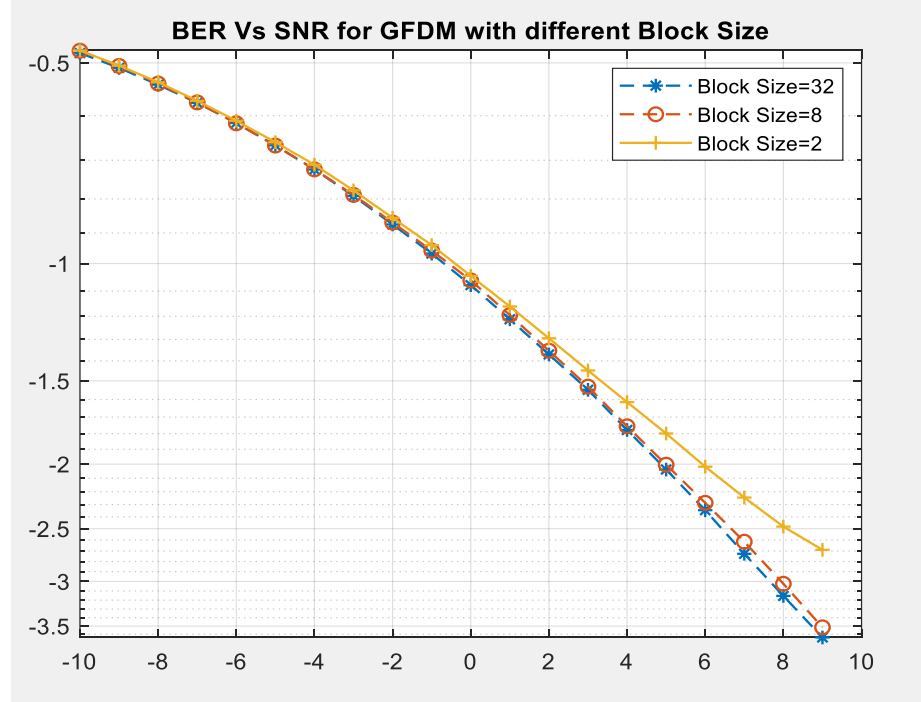
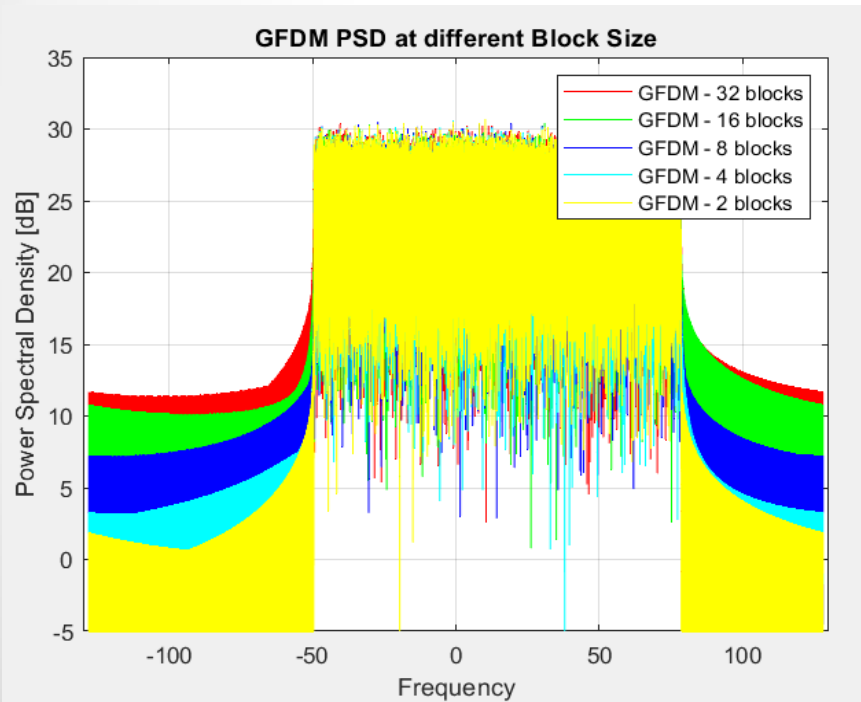
FBMC BER Performance with different Filter at 64QAM



System Simulation and Results

GFDm simulation

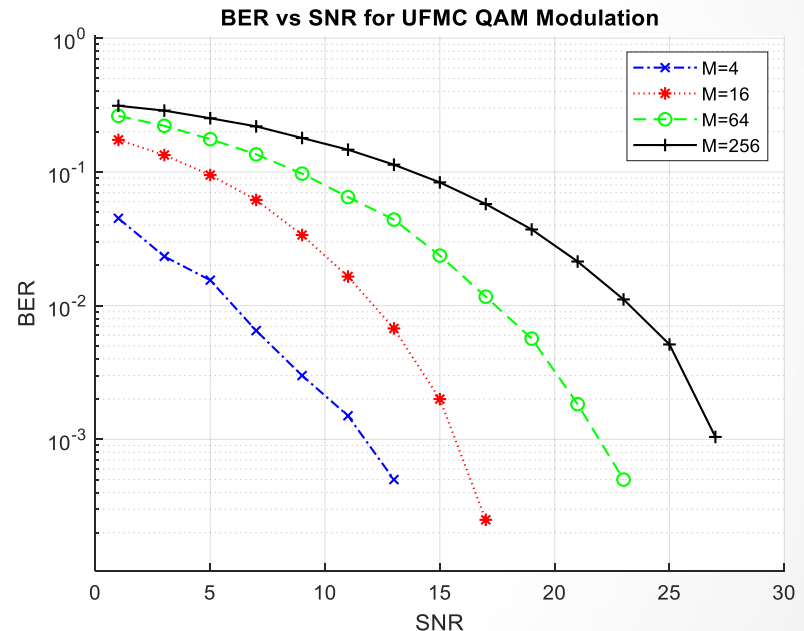
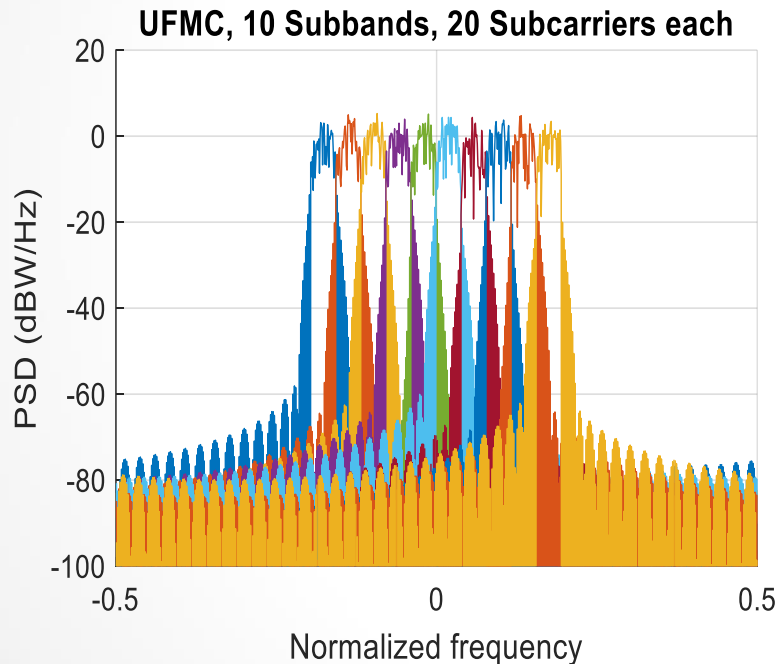
- ✓ PSD for GFDm signal with different block size is shown and it is observed that with increase in block size the PSD and BER performance decreases and spectral leakage increases.
- ✓ GFDm has the flexibility to choose the number of blocks.



System Simulation and Results

UFMC simulation

- ✓ The performance parameters PSD and BER are simulated.
- ✓ To achieve good frequency spectrum utilization, higher order QAM mapping technique is to be applied



Conclusion

- FBMC is the most spectrally efficient technique but has no backward compatibility with 4G and MIMO
- UPMC, F-OFDM and GFDM are compatible with 4g as well as most suitable for 5G but have higher complexity in system design.

S. No.	Metrics for comparison of waveforms	Performance report
1	Compatibility with 4G (LTE) and MIMO	UPMC and F-OFDM waveforms are compatible with LTE but FBMC and GFDM does not support backward compatibility
2	Latency	FBMC has highest latency and UPMC is used for low latency applications
3	Out of band emission (Spectral leakage)	Best filter performance and lowest OOB is in FBMC
4	Compatibility with high mobile applications	FBMC is the best suitable waveform for high mobile applications
5	Flexible to adapt in different scenarios	UPMC is the most flexible to with various numerology
6	Compatibility for Short burst traffic	UPMC and GFDM are most suitable for short burst traffic
7	Transceiver Design complexity	UPMC and GFDM has most complexity
8	Best spectrum utilization	F-OFDM uses scattered frequency band so saves bandwidth

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THANK YOU
Questions?